

18.11.2025(E)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: July 2025– November 2025
Maximum Marks: 50
Examination: End-Semester Examination
Duration: 2 Hrs.
Programme code: 03 & 41
Class: TY
Semester: V (SVU2023)
Programme: B.Tech EXTC / B.Tech CCE
Institute/School/Department: K. J. Somaiya School of Engineering
Name of the department/Section/Center:
 EXTC & CCE

Course Code: 216U03C502 & 216U41C502
Name of the Course: Computer Communication Networks
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any TWO	
i)	Explain the wired transmission media in computer networks	05
ii)	Explain w.r.t. OSI model, how do two hosts in the computer network communicate via router.	05
iii)	Give the classification of unicast routing protocols. Explain Link state routing.	05
Q.2	Attempt any ONE	10
i)	a) Explain the working of Selective Repeat protocol with the help of suitable diagram. b) State the classification of media access control (MAC) protocols. A CSMA/CD network has a bandwidth of 100 Mbps. If the maximum propagation delay is $51.2 \mu\text{s}$, determine the minimum frame size.	5 5
ii)	Generate the Hamming code for the data word 1011. Explain how is error detected at the receiver?	10
Q.3	Attempt any ONE	10
i)	What is the use of Address Resolution Protocol? Draw and explain the ARP Header format. Generate the ARP Request packet for the following network if router R1 generates the ARP request for the Host H1. <pre> graph LR R1[Router R1] --- H1[Host H1] R1 --- H2[Host H2] R1 --- H3[Host H3] </pre> Host H1: IP: 172.17.22.5, MAC: 00:00:0A:BB:28:FC Host H2: IP: 172.17.22.7, MAC: AA:10:0A:BB:28:2E Host H3: IP: 172.17.22.10, MAC: BB:12:5A:3B:C8:25 Router R1: IP: 172.17.22.1, MAC: 4C:1EB4:7B:2A:6F	10



ii)	a) An IPv4 network 176.11.0.0 is to be divided into 5 subnets. Determine the new subnet mask and address block of each subnet.	5
	b) Explain Error reporting messages in ICMP Protocol.	5
Q.4	Attempt the following	10
	a) Explain data transfer using 3-Way handshaking in TCP. State the flags used.	5
	b) What is the use of port numbers in transport layer. Explain various types of port numbers.	5
Q.5	Attempt the following	10
	a) What is the use of DNS? Explain, how it works?	5
	b) Write a note on HTTP.	5